

Chapter 18 — Studies in Biological Psychology

Student Notes Workbook · 学习笔记工作册

Raine (1997) · Brendgen (2005) · McDermott (2008) · Hoefelmann (2006)

Name 姓名: _____

Class 班级: _____

Start date: _____

How to use this workbook 使用方法

1. **Read one section** of the courseware, then fill in the matching workbook page. 边读课件边填空。
2. **Write out AMSPRC** for each study from memory after reading — Aim, Method, Sample, Procedure, Results, Conclusion.
3. **Check answers** on p. 13 and tick the tracker below. 对答案并打卡。每页右上角标有对应课件章节。

Syllabus map — what the exam can ask 大纲考点

Syllabus	Content 内容	Workbook pages
AMSPRC	Six-step structure: Aim · Method · Sample · Procedure · Results · Conclusion	p. 2
Classic study	CLASSIC Raine et al. (1997) — PET scan of NGRI murderers	pp. 3–5
Contemporary 1	CONTEMP Brendgen et al. (2005) — twin study of aggression	pp. 6–7
Contemporary 2	CHOICE A McDermott et al. (2008) — MAOA & hot-sauce or CHOICE B Hoefelmann et al. (2006) — sleep & lifestyle	pp. 8–9
Exam skills	AMSPRC answer writing · 2 / 4 / 8-mark structure · evaluation language	pp. 10–11
Self-test	Brain dump from memory · sentence patterns · answer key	pp. 12–13
Exam format 考试形式: Paper 3 — one 8-mark evaluate on Raine, one 7-mark on a contemporary study (procedure 3 + evaluate 4). Always include numbers and named examples (AO3).		

Review tracker 复习打卡

- ☐ First pass — all blanks filled 第一遍填完 Date: _____
- ☐ AMSPRC tables written from memory for all 4 studies 默写四篇研究的 AMSPRC Date: _____
- ☐ Answer key checked — corrections in pen 对过答案并用笔订正 Date: _____
- ☐ Brain dump page done from memory 合上笔记完成自测页 Date: _____

What AMSPRC means 含义填空

A = _____ (研究目的 Aim) — what were they trying to find out?
M = _____ (研究方法 Method) — quasi-experiment / twin / questionnaire / etc.
S = _____ (被试 Sample) — who took part, how many, how chosen
P = _____ (实验流程 Procedure) — what they did, step by step
R = _____ (结果 Results) — what they found (numbers!)
C = _____ (结论 Conclusion) — what they said it meant

(黄金规则：无论 2 分 / 4 分 / 8 分题，写答案时永远按 AMSPRC 的顺序。考官改卷也是按这个顺序对照 mark scheme 的。)

The four studies at a glance 四篇研究速览

Study	Focus	Method	Exam role
Raine 1997	Brain abnormalities in NGRI murderers	_____ (PET scan PET 扫描)	Classic — 8-mark evaluation
Brendgen 2005	Genes vs env in social & physical aggression	_____ (Twin study 双生子研究)	Contemporary — describe + evaluate
McDermott 2008	MAOA gene + provocation	_____ (Hot-sauce game 行为经济学)	Contemporary choice A
Hoefelmann 2006	Lifestyle & sleep in students	_____ (Questionnaire 问卷)	Contemporary choice B

Mark-scheme logic — how marks map to AMSPRC 分数与 AMSPRC 的对应

Question type	What you must do	AO1 (knowledge)	AO3 (evaluation)
2-mark describe	Pick any 2 of {Aim, Method, Sample, Procedure, Results, Conclusion}	2 marks	—
3-mark procedure	Just the Procedure step — 3 clear details	3 marks	—
4-mark evaluate	2 strengths/weaknesses, each with WHY (AO3)	—	4 marks
7-mark evaluate	(a) Procedure 3 marks + (b) 2 weaknesses with AO3 = 4 marks	3 marks	4 marks
8-mark evaluate	~4 evaluation points, each Point + Evidence + Why	1–2 marks	6–7 marks

Spot the AO3 trap 区分 AO1 和 AO3

AO1 = "Raine used a PET scan with 41 NGRI murderers." (just stating a fact)

AO3 = "Using only NGRI murderers means the sample is unrepresentative of all violent offenders, so findings cannot generalise." (fact + the implication)

每写一条 evaluation 必须有 WHY；只罗列名词（如 "low ecological validity"）拿不到满分。

One-line summary 一句话总结

To investigate whether brain activity in _____ (NGRI murderers 因精神错杀人的凶手) differs from a control group of matched non-murderers, using a _____ (PET scan PET 扫描) .

Sample breakdown — fill in the numbers 填空

	Experimental group 实验组	Control group 对照组
Who?	Murderers / _____ (manslaughter 误杀) , pleading NGRI	Non-murderers, _____ (matched 配对的) on age & gender
Total n	_____	_____
Males / females	_____ / _____	_____ / _____
Screened for	General health, physical exam, medical history, psychiatric interview	Excluded if history of _____ , _____ (head trauma 头部创伤) or _____

Conditions inside the experimental group 实验组内部构成

Diagnosis	How many	Translation 中文
Schizophrenia	_____	精神分裂症
Organic brain damage / head injury	_____	器质性脑损伤/头部创伤
Substance abusers	_____	物质滥用者
Affective disorder	_____	情感障碍
Epilepsy	_____	癫痫
Hyperactivity / learning disability	_____	多动/学习障碍
Passive-aggressive / paranoid personality disorder	_____	被动攻击型/偏执型人格障碍

(提示：加起来应该等于 41 —— 核对一遍，错的就翻回课件 §2 改。)

Standardisation — what was the same for everyone? 标准化 (填空)

All participants remained _____ (medication-free 不用药) for _____ weeks before the PET scan.
This controls for the effect of _____ (medication 药物) on brain activity.

Exam practice 自己写 2 分 sample 题

"Describe the sample used by Raine et al. (1997)." (2)

写两句话，覆盖：① 每组人数 + 性别 + 配对 ② 实验组的 NGRI 身份（或对照组的排除标准）。

What the PET scan actually measured PET 在量什么

PET = **Positron Emission Tomography** (正电子发射断层扫描) . It measures the _____ (metabolic rate 代谢率) of different brain areas using a _____ (radioactive tracer 放射性示踪剂) (FDG). Active brain cells take up more glucose → more tracer → _____ (more signal 更多信号) .

Put the procedure in order — number the steps 1–5 排序

Order	Step
_____	_____ (CPT 持续执行任务) for 10 minutes — keeps attention active so brains are in the same state.
_____	Practise the _____ on a sequence of blurred numbers.
_____	Inject the radioactive glucose tracer called _____ (FDG 氟脱氧葡萄糖) .
_____	Continue on the CPT for another _____ minutes.
_____	Perform the _____ (PET scan PET 扫描) to measure metabolic rate.

(提示：答案在右边的 hint 区域里——但先自己排，再揭开看是否一致。)

Why each step matters — match the reason 连线练习

Step	Why it was done 目的
10-minute CPT practice	① So the brain is in a stable state during the scan → can compare _____ (fairly 公平地)
FDG injection	② Radioactive tracer lights up _____ (active brain cells 活跃脑细胞)
32-minute wait	③ Gives FDG time to be _____ (taken up 被吸收) by brain cells
PET scan	④ Measures _____ (metabolic rate 代谢率) (activity level) in each brain area

Operationalisation — turn the concept into a number 把概念变成可测量的变量

The DV (因变量) was the _____ (metabolic rate 代谢率) in each brain area, measured in _____ (FDG uptake FDG 摄入量) units. The IV (自变量) was _____ (group membership 组别) : murderer vs control.

Exam sentence builder 必背句式

"Participants were injected with FDG, a radioactive glucose analogue. Active brain cells absorb more FDG, so the PET scan indicates which brain areas are _____ by showing where tracer _____ (accumulates 累积) is highest." (4 marks)

Direction of effect 方向填空

Murderers showed **lower** activity in some areas, **higher** activity in others, and **identical / asymmetric** activity in yet others. The key brain area for self-control is the _____ (prefrontal cortex 前额叶皮层) .

Brain-area results table — fill in direction 填方向

Brain area	Direction in murderers	What it might mean
Prefrontal cortex (前额叶皮层)	_____	Impulsivity, poor self-control, can't learn from consequences
Parietal areas (顶叶) (incl. left angular gyrus)	_____	_____ (attention 注意 & language 语言)
Occipital lobe (枕叶)	_____	Visual processing — less directly linked to violence
Temporal lobe (颞叶)	_____	No difference found
Corpus callosum (胼胝体)	_____	Reduced _____ (communication 沟通) between hemispheres
Amygdala (杏仁核)	_____	Emotion & _____ (fear 恐惧) regulation
Hippocampus (海马)	_____	Learning & _____ (memory 记忆)
Thalamus (丘脑)	_____ ((right side))	Sensory relay, _____ (arousal 唤醒)

(填法提示：① Lower / Higher / Identical / Asymmetric。② Asymmetric = left 与 right 不一样 (一边高一边低))

Conclusion in one sentence 结论一句话

Raine concluded that NGRI murderers showed brain dysfunction in areas linked to _____ (impulsivity 冲动) and _____ (emotion regulation 情绪调节) . However, this is only a _____ (correlation 相关) — it does not prove brain differences _____ (cause 导致) murder.

✓ STRENGTHS 优点 (写关键词)

- **Large sample:** 41 NGRI murderers is _____ (large for this population 对这类人群算大) .
- **Matched controls:** same age & gender → improves _____ (internal validity 内部效度) .
- **Standardised:** same CPT + PET → _____ (objective 客观) comparison.

✗ WEAKNESSES 缺点

- **Unrepresentative:** NGRI murderers are _____ (atypical 非典型的) violent offenders.
- **Correlation:** lower brain activity could be the _____ (result 结果) of violence, not the cause.
- **Ethics:** PET exposes controls to _____ (radiation 辐射) they didn't clinically need.

Twin logic 核心逻辑 (必背)

MZ (identical, 同卵) share _____ (100%) of genes; DZ (non-identical, 异卵) share about _____ (50%) . If MZ twins are more _____ (similar 相似) on a trait than DZ twins, _____ (genes 基因) are likely involved. If $MZ \approx DZ$, the trait is more _____ (environmental 环境) .

Sample breakdown 样本构成 (填空)

Source	Detail
Where did they come from?	_____ (Quebec Newborn Twin Study (QNTS))
When were they born?	November _____ – July _____
Initial sample	_____ twin pairs tested
Final sample with complete data	_____ (234 pairs)

Breakdown of the 234 pairs — fill the 5 cells 五组样本构成

Sub-group	Pairs	Genetic overlap 共享基因
MZ males	_____	_____ (100%)
MZ females	_____	_____ (100%)
DZ males (same sex)	_____	_____ (~50%)
DZ females (same sex)	_____	_____ (~50%)
DZ mixed-sex	_____	_____ (~50%)
Total	234	—

Procedure — what they did 流程

- **Longitudinal design:** data collected at _____ (5, 18, 30, 48, 60 months) + again at age _____ (6) (focus).
- **Two rating sources:** _____ (teacher 教师评分) ratings (spring term) + _____ (peer 同伴评分) ratings by classmates.
- **Scales used:** PSBS-T (Crick et al., 1997) + Direct & Indirect Aggression Scales (Björkvist et al., 1992).
- **Item scoring:** _____-point scale: 0 = never, 1 = sometimes, 2 = _____ (often 经常) .

Why longitudinal matters 为什么是 longitudinal

Collecting data at multiple time points lets researchers track _____ over development, instead of just one snapshot. But it also means more participants drop out — here, 322 → _____ pairs (about _____ % lost).

Two types of aggression 两种攻击行为**Physical aggression** (身体攻击) = hitting, fighting, kicking.**Social aggression** (社会性攻击) = ignoring others, spreading rumours, threatening to withdraw friendship.Can be **overt** (公开的) or **covert** (隐蔽的) .**MZ vs DZ correlations — fill the conclusion** 填空

Type of aggression	MZ similarity	DZ similarity	Conclusion
Physical aggression	_____ (much higher 远高于)	lower	_____ (strong genetic component 强遗传成分)
Social aggression	_____	_____	Better explained by _____ (shared environment 共同环境)
Physical–social correlation	Correlation found, best explained by _____		General tendency to be aggressive may be _____ (genetic 遗传) , but <i>expression</i> is environmental

Direction of influence 方向填空

Physical aggression may _____ (lead to 导致) **social aggression**, but _____ (not the other way round 反过来不会) .

Reason: as children grow, they develop _____ (language/cognitive skills 语言与认知能力) and learn more _____ (socially acceptable 社交上可接受的) ways to express the same aggressive tendency.

Evaluation 评价 (填空)✓ **STRENGTHS**

- **Two rating sources** (teacher + peer) → _____ (triangulation 三角互证) increases validity.
- **Longitudinal** data from _____ (early childhood 婴幼儿期) to age 6.
- **Large sample**: 234 twin pairs.

✗ **WEAKNESSES**

- **Small sub-groups** when split by sex → hard to _____ (generalise 推广) .
- **Peer ratings** by 6-year-olds may be _____ (unreliable 不可靠) .
- **Zygosity check**: not all twins had a _____ (DNA test DNA 检测) .

Exam sentence builder 必背句式

"Brendgen et al. (2005) found that MZ twins were _____ more alike than DZ twins on teacher ratings of physical aggression, suggesting a _____ influence. For social aggression, MZ and DZ similarities were _____, suggesting it is more _____."

MAOA in one sentence 一句话

The **MAOA gene** (单胺氧化酶 A 基因) breaks down neurotransmitters like serotonin. There are two variants: **MAOA-L** (low activity, _____ (higher aggression 更高攻击性)) and **MAOA-H** (high activity, lower aggression).

Sample — fill in the key numbers 关键数字

Item	Detail
Participants	_____ male college students
Why only males?	MAOA is on the _____ (X chromosome X 染色体) — harder to assign activity in females
MAOA-L carriers	_____ (27%) of the sample
MAOA-H carriers	remaining ~ _____ (73%)

Procedure — order the 3 phases 排序

Order	Phase	What happens
_____	Quiz phase	5 vocab questions; 20 pts each; 10 pts = US\$ _____; max = US\$10
_____	Stealing phase	"Opponent" steals _____ (80%) (high take) or _____ (low take) of earnings
_____	Punishment phase	Force "opponent" to drink _____ (hot sauce 辣椒酱) — DV = amount allocated

(重要：所谓"opponent"其实根本不存在，是电脑生成的——这是 deception (欺骗)。每个被试玩 4 轮，每轮一个"对手"。)

Results — fill in the %s (look at the figure embedded in the courseware §4)

Condition	MAOA-L	MAOA-H	Interpretation
80% stolen — gave some hot sauce	_____ %	_____ %	MAOA-L more aggressive under high provocation
20% stolen — gave some hot sauce	_____ %	_____ %	Smaller difference under low provocation
80% stolen — gave maximum hot sauce	_____ %	_____ %	_____ (Strong effect 大效应) under high provocation
20% stolen — gave maximum hot sauce	_____ %	_____ %	Weak effect under low provocation

(提示：80% stolen 时的 $p < ______$ (significant), 20% stolen 时 $p = ______$ (not significant) 。)

Conclusion — gene × environment interaction

MAOA-L individuals are *not* automatically aggressive. They become more aggressive **only when provoked**. This is a _____ (gene–environment interaction 基因-环境交互) : genes + provocation together produce the effect, neither alone.

Aim in one sentence 目标一句话

To investigate which _____ (lifestyle factors 生活方式因素) are associated with the quality and _____ (duration 时长) of sleep in high-school students.

Sample — fill in the details

Item	Detail
Originally selected	~ _____ students from _____ schools in Brazil
Cities	_____ (Florianópolis 南部) & _____ (Recife 北部)
Age range	_____ - _____ years
Type of school	_____ (night classes 夜校) at public schools
Completed both assessments	_____ students

Two research designs — fill the gaps 两种设计

Design	When it was used	What it compared 比较什么
Cross-sectional (横断)	March _____	Variables at _____ (a single time point 单一时间点)
Prospective (前瞻/纵向)	March & December _____	Same participants _____ (over 9 months 9 个月内)

Results — fill the 4 cells 四个结果

Measure	Cross-sectional	Prospective	Key finding
Poor sleep quality	_____ %	_____ %	High & stable
Good sleep quality	_____ %	_____ %	High & stable
Sleep < 8 hours	_____ %	_____ %	Most had _____ (insufficient sleep 睡眠不足)
Sleep ≥ 8 hours	_____ %	_____ %	Only a minority had _____ (sufficient sleep 充足睡眠)

(提示: cross-sectional 看到 lifestyle ↔ sleep 有关联, 但 prospective 9 个月没有证实 → correlational, not causal.)

Cross-sectional vs prospective — what changed? 设计差异导致结果差异

Cross-sectional found associations (e.g. higher _____ (snack consumption 零食摄入) → poor sleep).

Prospective found **none** of these held up over 9 months. This means the cross-sectional links are probably _____ (correlational 相关的), not _____ (causal 因果的).

Question (8 marks): Evaluate Raine et al.'s (1997) study of brain activity in murderers pleading NGRI. In your answer you should consider the strengths and weaknesses of the study.

Plan first — 4 evaluation points × (Point + Evidence + Why) 4 个评价点框架

#	Point 评价点	Evidence from Raine 证据	Why (AO3) 为什么是优/缺点
1	_____ (strength 优点)	_____	_____
2	_____	_____	_____
3	_____ (weakness 缺点)	_____	_____
4	_____	_____	_____

Write the 8-mark answer on the lines below 用下面的空格写完整答案

Method marks 方法分提醒

考官按段落给分：① 一段 AO1 简要描述 (study + sample) → 1–2 分 ② 每段 Point + Evidence + Why → 1 分 ③ 全篇 ~4 段 AO3 + conclusion → 余下分数。
每段至少三句话 (POE 结构) 才稳拿满分；只写 "low ecological validity" 这种关键词不算 AO3。

Self-mark 自评对照

- ☐ AO1: 1–2 句描述 (含 41+41 + PET + NGRI)
- ☐ AO3: 2 strengths × POE (large sample / matched / standardised)
- ☐ AO3: 2 weaknesses × POE (unrepresentative / correlation / ethics)
- ☐ Conclusion: "useful evidence but cannot establish causation"

During your studies of biological psychology, you will have learned about one of the following contemporary studies: McDermott et al. (2008) or Hoefelmann et al. (2006).

(a) Describe the procedure of your chosen study. (3)

(b) Explain two weaknesses of your chosen contemporary study in terms of validity. (4)

Pick one 选一个 — circle or tick

☐ McDermott (2008) — MAOA & hot sauce ☐ Hoefelmann (2006) — sleep & lifestyle

(a) Describe the procedure (3 marks) — write 3 clear steps

(b) Two weaknesses of validity (4 marks) — POE × 2

#	Weakness	Why it reduces validity (AO3)
1	_____	_____
2	_____	_____

If you picked McDermott — 三个可选 weakness 提示

- ① **Low ecological validity:** forcing an unseen opponent to drink hot sauce is not real-life aggression → findings don't generalise to everyday behaviour.
- ② **Deception:** participants believed there was a real opponent → ethical issues, task feels artificial.
- ③ **Sample bias:** male college students only → cannot generalise to females or non-students.

If you picked Hoefelmann — 三个可选 weakness 提示

- ① **Self-report bias:** sleep & lifestyle all self-reported → may be inaccurate (social desirability / poor recall).
- ② **Night-school sample:** only night-class students → not generalisable to day-school students.
- ③ **Correlational:** cannot establish causation between lifestyle & sleep (cross-sectional links did not replicate in prospective analysis).

- AMSPRC — what does each letter stand for?

- Raine — sample (n, gender, matching, exclusions):

- Raine — procedure in 4 steps from CPT to PET:

- Raine — 4 key brain areas and what was found (lower/higher/asymmetric):

- Brendgen — MZ vs DZ logic (n, similarity rule, what each implies):

- Brendgen — physical vs social aggression findings + direction:

- McDermott — MAOA-L vs MAOA-H + the 3-phase hot-sauce procedure:

- McDermott — 4 result cells (80%/20% stolen × some/max hot sauce):

- Hoefelmann — sample (n, cities, age, type of school) + two designs:

- Hoefelmann — why did cross-sectional results NOT replicate prospectively?

Exam sentences to memorise 背下来的句式

When asked...	Sentence pattern 句式
Describe the sample (2)	"The sample consisted of _____ [group A] and _____ matched controls; participants were screened for _____ and remained _____."
Describe the procedure (3)	"Participants first _____, then were given _____, and finally _____; the DV was _____."
Interpret twin data (Brendgen)	"MZ twins were _____ more similar than DZ twins on _____, suggesting a _____ influence; but MZ ≈ DZ on _____ → _____."
Evaluate validity (4)	"A weakness is _____ because _____, which means _____."

Final self-check 最后自问

合上笔记,能否在 10 分钟内默写出: ① Raine 41+41 + CPT + FDG + 32 分钟 + PET? ② Brendgen MZ ≈ DZ (social) vs MZ > DZ (physical)? ③ McDermott 80%-stolen L=75%/H=62%? ④ Hoefelmann 989 学生 + 54.3% 好眠 + 76.7% 不足? 卡住就回课件对应章节重看。

p.1 — 怎么用: 读一节填一页 · 默写 AMSPRC · 对答案打卡. 大纲: AMSPRC p.2 / Raine pp.3–5 / Brendgen pp.6–7 / McDermott p.8 / Hoefelmann p.9 / Exam pp.10–11 / Self-test pp.12–13.

p.2 — Aim 研究目的 · Method 方法 · Sample 被试 · Procedure 流程 · Results 结果 · Conclusion 结论. 四研究: PET scan / Twin study / Hot-sauce game / Questionnaire. 经典 Raine · 当代 Brendgen · 选 McDermott 或 Hoefelmann.

p.3 — NGRI · PET · 41 · 41 · 39 males / 2 females (both groups) · matched · seizures · head trauma · substance misuse · 内部构成: 6 (schizophrenia) · 23 (organic brain damage / head injury) · 3 (substance abusers) · 2 (affective disorder) · 2 (epilepsy) · 3 (hyperactivity / learning disability) · 2 (passive-aggressive / paranoid) → 加起来 = 41. medication-free · 2 weeks · medication. 2 分 sample 题答案: 一句话人数 + 配对, 一句话 NGRI / 排除.

p.4 — 顺序: ② practise CPT (10 min) → ① CPT (10 min) → ③ FDG injection → ④ CPT another 32 min → ⑤ PET scan. 目的连线: practice → fairly; FDG → active brain cells; 32-min wait → taken up; PET → metabolic rate. DV = metabolic rate / FDG uptake. IV = group membership.

p.5 — Prefrontal: lower. Parietal: lower · attention & language. Occipital: higher. Temporal: identical. Corpus callosum: lower · communication between hemispheres. Amygdala: asymmetric · fear. Hippocampus: asymmetric · memory. Thalamus: higher (right) · arousal. 结论: impulsivity · emotion regulation · correlation · cause. 优点: large for this population · internal validity · objective. 缺点: atypical · result · radiation.

p.6 — 100% · 50% · similar · genes · environmental. 来源: Quebec Newborn Twin Study (QNTS). 出生: 1995 – 1998. 初始: 322. 完整: 234. 5 子组: 44 MZ males · 50 MZ females · 41 DZ males · 32 DZ females · 67 DZ mixed-sex · 加起来 = 234. 时间点: 5, 18, 30, 48, 60 months · age 6. 评分源: teacher · peer. 量表: PSBS-T + Direct & Indirect Aggression Scales. 计分: 3-point · 0=never · 1=sometimes · 2=often. 纵向: development · 234 · 27% lost.

p.7 — 身体攻击: much higher (MZ) → strong genetic component. 社会攻击: similar (MZ ≈ DZ) · similar → shared environment. 相关: genetic · genetic · environmental. 方向: lead to · not the other way round · language/cognitive skills · socially acceptable. 优点: triangulation · early childhood · 234. 缺点: generalise · unreliable · DNA test.

p.8 — MAOA-L · higher aggression. 样本: 78 · X chromosome · 27% · 73%. 顺序: ① Quiz · ② Stealing · ③ Punishment. 偷窃: 80% / 20%. 辣椒酱 · DV = amount. 4 cells: 75 / 62 · 40 / 34 · 44 / 19 · 12 /

6. 显著性: 0.01 (significant) · 0.19 (not significant). 结论: gene-environment interaction.

p.9 — lifestyle factors · duration. 样本: ~2000 · 20 · Florianópolis · Recife · 14–24 · night classes · 989. 横断: 2006 · a single time point. 前瞻: 2006 · over 9 months. 4 cells: 45.7% / 45.8% · 54.3% / 54.2% · 76.7% / 77.5% · 23.3% / 22.5% · insufficient sleep · sufficient sleep. 关联: snack consumption · correlational · causal.

p.10 — 4 evaluation points: ① Strength: large sample for this population — 41 NGRI murderers with matched controls improve validity. ② Strength: standardised procedure (CPT + PET) gives objective data. ③ Weakness: unrepresentative sample — NGRI murderers have unusual characteristics (mental illness / head injury), so cannot generalise. ④ Weakness: correlation not causation — lower brain activity could be the result of violence or substance use, not the cause. 5: ethics — radiation exposure to controls. 收尾: useful evidence but cannot establish causation.

p.11 — McDermott: (a) 三步: ① Participants answered 5 vocab questions to earn points (10 pts = \$1). ② A computer-generated "opponent" stole 20% or 80% of earnings. ③ Participants could punish the opponent by allocating hot sauce; DV = amount of hot sauce; 4 rounds. (b) ① Low ecological validity: forcing an unseen opponent to drink hot sauce is not like real-life aggression → cannot generalise. ② Deception: participants believed there was a real opponent → ethically problematic + task feels artificial. Hoefelmann: (a) 三步: ① Students in 20 Brazilian schools completed a questionnaire in March 2006. ② Same students completed the questionnaire again in December 2006. ③ Questions measured sleep quality ("How often do you think you sleep well?" 5-point scale), sleep duration (hours/day), and lifestyle factors (activity / snacks / soft-drink / TV). (b) ① Self-report bias: all data self-reported → may be inaccurate (social desirability, poor recall) reducing validity. ② Night-school sample: only night-class students → not generalisable to day-school students (population validity).

p.12 — 背句模板填空答案: ① 41 · 41 · general health / head trauma · medication-free ② did X · Y · Z · DV = metabolic rate / hot sauce amount / sleep score ③ significantly · physical aggression · genetic · social aggression · environment ④ X · evidence · implication. Key numbers: Raine 41+41 / 39M+2F / 23 brain damage / 6 schizophrenia; Brendgen 234 pairs / 5 sub-groups / MZ ≈ DZ social / MZ > DZ physical; McDermott 78 male / 27% MAOA-L / 75 vs 62 / 44 vs 19; Hoefelmann 989 / 20 schools / 54.3% good / 76.7% under 8h.

最后提醒: Ch18 的核心是 AMSPRC——先把每个研究按 A/M/S/P/R/C 六字诀背一遍, 再去记具体数字。每次写答案前先列关键词, 再串成句。考官找的是 POE 三层结构 (Point + Evidence + Why), 缺一不可。